Business requirements document template

Project name: 30-DayReadmission Risk Dashboard

Project manager: Dr. AmandaClark

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Document status: ☑ Draft ☐ Proposed ☐ Validated ☐ Approved

1. Executive summary

This initiative aims to analyze patient encounter data from 100,000+ diabetic patient records to identify key factors driving 30-day hospital readmissions. By leveraging Excel, SQL, and Power BI, ReadmitGuard Analytics will quantify readmission risk factors across demographics, diagnoses, and medications. The resulting interactive dashboard will support healthcare teams in proactive decision-making, enabling early identification of high-risk cohorts and data-driven interventions to reduce CMS readmission penalties

2. Project objectives

1. Identify and quantify 30-day readmission risk factors across demographics, diagnoses, and medications with 95% data accuracy.
2. Visualize readmission distribution by age group, gender, race, diagnosis (diag_1), and payer code using interactive Power BI dashboards.
3. Reduce manual reporting effort by 40% through automated Excel pivot reports and SQL aggregations.
4. Deliver evidence-based intervention recommendations for the top 3 high-risk patient cohorts.
5. Enable compliance reporting to support reduction of CMS readmission penalties.

3. Project scope

In Scope:
-Analysis of historical diabetic_data.csv (encounter_id, demographics, diagnoses, medications, readmitted flag)
-Data cleaning and transformation in Excel and SQL (schema: healthcare)
-SQL analysis of readmissions, length of stay(LOS), admission sources
-Power BI dashboards with AI Quick Measures
-Role-specific views for clinicians and admins.

Out of Scope:
-Real-time EHR or hospital system integration, Clinical decision-making
-Predictive ML model deployment beyond descriptive analytics, PHI re-identification or external data linkage

4. Business reqirements

Priority level	Critical level	Requirement description
High	Critical	BR1: Consolidate patient encounter data into cleaned SQL t
High	Critical	BR2: Generate readmission trend reports via Excel pivot tab
High	Critical	BR3: Build interactive Power BI dashboard with 5 visuals (di
Medium	High	BR4: Apply AI summaries using Excel =AI.TABLE and Powe
Medium	High	BR5: Create role-based filters for Doctor and Admin views

5. Key stakeholders

Name	Job role	Duties
Dr. Amanda Clark	Project Manager	Approves BRD, owns timeline and deliverables
Hospital Administrator	Operations Lead	Uses dashboards for resource planning
Compliance Officer	Governance	Ensures HIPAA-compliant anonymized data use
BI Analyst	Analytics	Builds SQL schema and Power BI visuals
Business Analyst	Execution	Leads analysis, Excel exploration, reporting

6. Project constraints

Constraint	Description
Data	Limited to anonymized 1999-2008 diabetic encounters; no longitudinal history be
Tools	Restricted to Excel, SQL, Power BI Desktop only
Privacy	No access to live PHI or identifiable patient data
Timeline	Delivery aligned to June 2026 quarterly reporting deadline

7. Cost-benefit analysis

Cost	Benefit
Analyst time (80 hrs @ $75/hr = $6,000)	$15,000 saved per avoided 30-day readmission penalty
Training on AI features ($500)	40% reduction in manual reporting (20 hrs/month = $18,000/yr saved)
Existing licenses ($0 incremental)	40% reduction in manImproved resource allocation and faster high-risk identifica
Documentation & overhead ($400)	Data-driven policy decisions and compliance support
Total cost: $6,900	**Expected ROI:** 320% in Year 1 (based on 2 prevented readmissions + repo